Light as an EM wave

Derivation of wave equation from Maxwell's equations, in a vacuum and in clear dialectrics

Typo in Steck Eq 4.16

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$
$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$
$$\vec{\nabla} \cdot \vec{B} = 0$$
$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

In a vacuum we set $\rho = 0$ (charge density) and $\vec{J} = 0$ (current density), but in a glass this won't be true. In a vacuum, we can derive

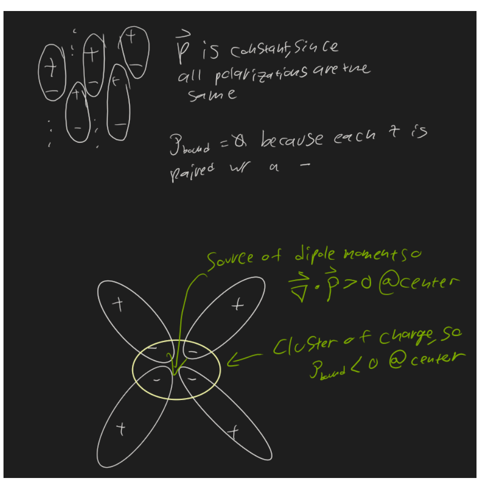
$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$
$$\implies c_0 = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Lets look inside a crystal. We can set a "free" and a "bound" charge density. We also define $\vec{P}$ - the "polarization density", the electric dipole moment per unit volume.

$$\vec{\nabla} \cdot \vec{E} = \left(\frac{\rho_{\text{free}}}{\epsilon_0} + \frac{\rho_{\text{bound}}}{\epsilon_0} \right)$$
$$\epsilon_0 \vec{\nabla} \cdot \vec{E} - \rho_{\text{bound}} = \rho_{\text{free}}$$

If we let $\vec{\nabla} \cdot \vec{\rho} = -\rho_{\text{bound}}$ then we get

$$\epsilon_0 \vec{\nabla} \cdot \left(\vec{E} + \frac{\vec{\rho}}{\epsilon_0} \right) = \rho_{\text{free}}$$



$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}_{\text{free}} + \mu_0 \vec{J}_{\text{magnetic}} + \mu_0 \frac{\partial \vec{\rho}}{\partial t} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

We can rearrange the right two terms to be

$$\mu_0 \frac{\partial}{\partial t} (\vec{\rho} + \epsilon_0 \vec{E}) = \mu_0 \frac{\partial \vec{D}}{\partial t}$$

Imagine a capacitor, two plates with free space between them. We define $\vec{D}$ as the electric field that would be in that free space. If we place a dielectric between the plates, then the charges in the material rearrange a bit and make the new overall $\vec{E}$.

We now have

$$\vec{\nabla} \times \vec{B} - \mu_0 \vec{J}_{\text{mag}} = \mu_0 \vec{J}_{\text{free}} + \mu_0 \frac{\partial \vec{D}}{\partial t}$$

Lets make this easier to solve by rewriting $\vec{j}_{\text{mag}}$ as a curl so that we can pull things together

$$\text{let } \vec{\nabla} \times \vec{M} \equiv \vec{J}_{\text{mag}}$$

$$\vec{\nabla}\left(\frac{\mu_0 \vec{B}}{\mu_0} - \mu_0 \vec{M}\right) = \mu_0 \vec{J}_{\text{free}} + \mu_0 \frac{\partial \vec{D}}{\partial t}$$

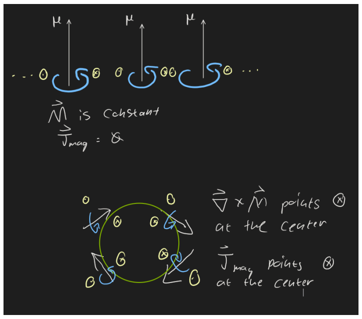
Lets get rid of all those pesky μ_0

$$\text{let } \vec{H} \equiv \frac{\vec{B}}{\mu_0} - \vec{M}$$

Then,

$$\vec{\nabla} \times \vec{H} = \vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t}$$

What is $\vec{M}$?
The units of $\vec{\nabla} \times \vec{M}$ are $\frac{\text{current}}{\text{distance}^{2}}$
So $\vec{M}$ has units $\frac{\text{current} \cdot \text{distance}^{2}}{\text{distance}^{3}}$
$\vec{M}$ is the "magnetization density", is the $\frac{\text{Magnetic Dipole Moment}}{\text{volume}}$



Summary

We can't yet derive the speed of light, but we are closer.
We now have new versions of Maxwell's equations.

$$\vec{\nabla} \cdot \vec{D} = \rho_{\text{free}}$$
$$\vec{\nabla} \cdot \vec{B} = -$$
$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$
$$\vec{\nabla} \times \vec{H} = \vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t}$$

We have polarization density $\vec{P}$

$$\vec{\nabla} \cdot \vec{P} = -\rho_{\text{bound}}$$
$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

As well as magnetization density $\vec{M}$

$$\vec{\nabla} \times \vec{M} = \vec{J}_{\text{magnetization}}$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$$